



SS1 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 1:

LESSON NOTE: WEEK 1 - ELECTRICAL CONTINUITY TESTING

1. DEFINITION OF ELECTRICAL CONTINUITY

Electrical continuity refers to the uninterrupted flow of electric current through a conductive path or circuit.

A conductor or component is said to be **continuous** if it forms a complete, unbroken path for electrons to flow from one terminal to another.

A **discontinuity** (such as a cut wire, broken fuse, or faulty switch) interrupts the circuit, resulting in zero or no current flow.

Continuity Testing: This is a diagnostic procedure used to verify the presence or absence of a complete path for current flow within an electrical or electronic circuit.

2. ELECTRICAL PRINCIPLE BEHIND CONTINUITY TESTING

The concept is based on **Ohm's Law**: $V = IR$

In continuity testing, the goal is to determine if resistance (R) is low enough (close to 0 ohms) to allow current to pass.

A **low or zero resistance** implies continuity (closed path), while a **very high or infinite resistance** implies an open circuit (discontinuity).

In continuity mode, a multimeter sends a small voltage/current across the circuit and detects whether a return path exists

3. TOOLS USED FOR CONTINUITY TESTING

(i) Digital Multimeter:

Most reliable instrument for testing.

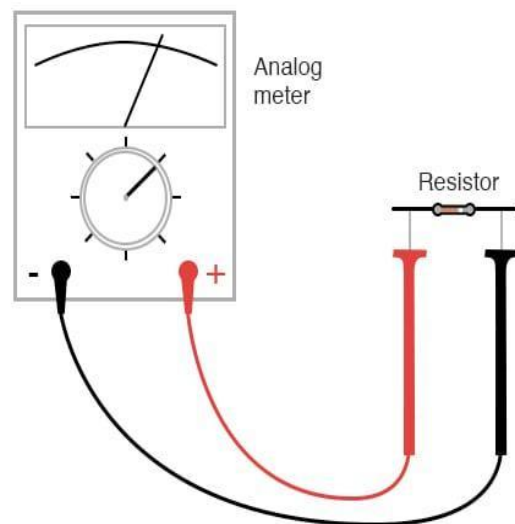
In continuity mode, it produces a beep if continuity exists.

Can also be used to measure resistance to determine continuity indirectly.



(ii) Analog Multimeter:

Uses a needle to display low resistance values
 Absence of deflection means no continuity.

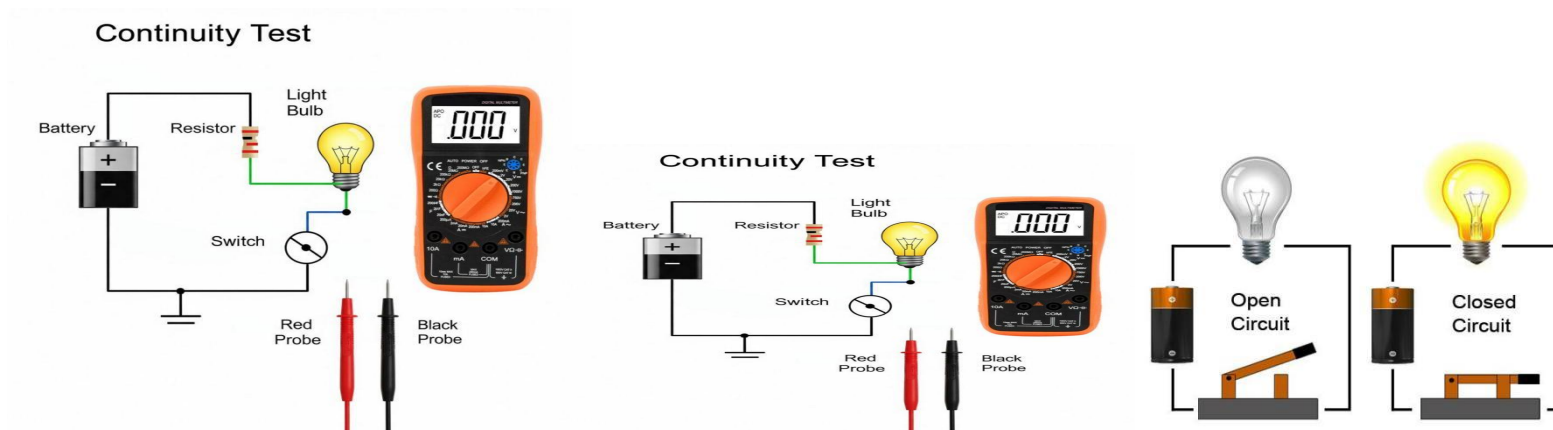


(iii) Continuity Tester:

A simple handheld tool with a built-in battery and bulb or buzzer.
 Lights up or buzzes when the circuit is complete.

(iv) Test Lamp:

Basic tool made from a bulb, battery, and two leads.
 Lights up if current flows through the path (i.e., continuity exists).



4. PROCEDURE FOR CONTINUITY TESTING (USING MULTIMETER)

Safety First:

Always turn off the power supply to the circuit/device before testing.

Steps:

Set the multimeter to continuity mode (or lowest ohm range if not available).

Insert the probes into the multimeter (black to COM, red to Ω port).

Touch the two probes together to confirm the meter is working (it should beep).

Touch the probes to both ends of the component or wire being tested.



Observe the reading or listen for the beep.

Interpretation:

Reading/Signal

Meaning

0 to $<1\Omega$ / beeping sound

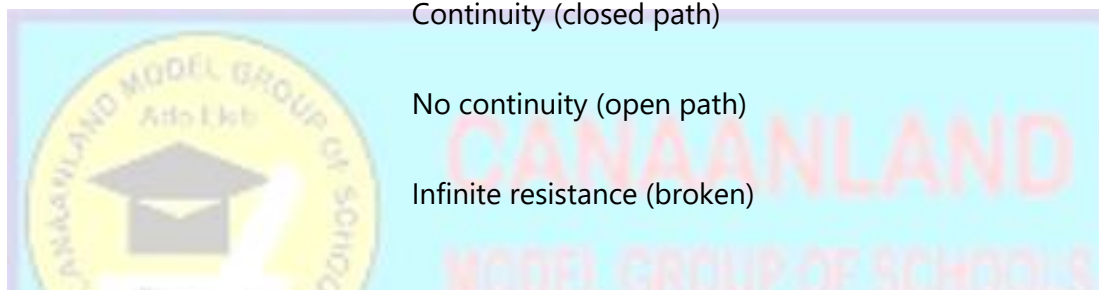
Continuity (closed path)

$>1000\Omega$ / No beep

No continuity (open path)

"OL" or " ∞ "

Infinite resistance (broken)



5. COMPONENTS THAT CAN BE TESTED FOR CONTINUITY

Electrical wires (live, neutral, earth)

Fuses (to check if blown)

Switches (open/close contact verification)

Circuit breakers

Conductive tracks on PCB

Coils in motors or transformers

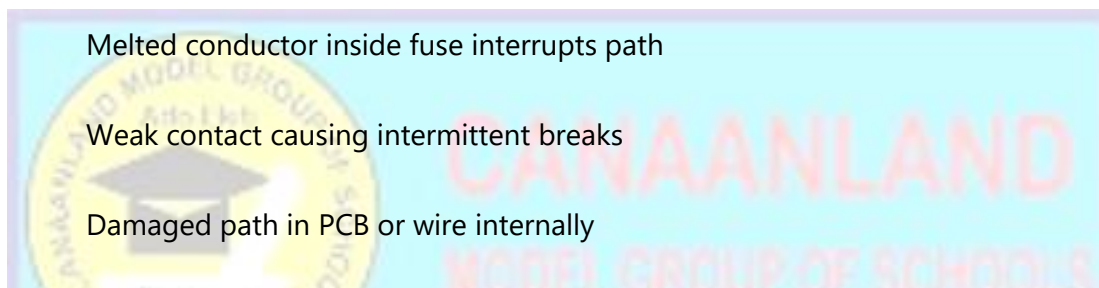
Connectors or terminals



6. COMMON FAULTS IDENTIFIED THROUGH CONTINUITY TESTING



Fault Type	Description
Open Circuit	Disconnected wire/component; no current flow
Short Circuit	Unintended low-resistance path
Blown Fuse	Melted conductor inside fuse interrupts path
Loose Connector	Weak contact causing intermittent breaks
Broken Trace	Damaged path in PCB or wire internally



7. APPLICATIONS OF CONTINUITY TESTING

Electrical Installation: Verifying wiring connections in homes/offices.

Automotive Repair: Checking wiring harnesses, sensors, and ignition paths

Electronics Production: Quality control in circuit boards.

Maintenance: Locating faults in old electrical systems.

Industrial Control Systems: Ensuring proper signal flow in control panels.



Aerospace & Rail Systems: Safety-critical integrity verification.

Instrument Calibration: Confirming connections before power-up

8. SAFETY PRECAUTIONS

Never perform continuity testing on a powered or live circuit

Ensure capacitors are discharged before testing

Use correct range settings on the multimeter.

Inspect probes for damage before use.

Handle tools with dry hands and rubber soles.

Always label tested wires to avoid confusion

Keep flammables away when testing.

9. CALCULATIONS AND PHYSICS APPLICATIONS

While continuity testing is primarily qualitative (pass/fail), it may involve resistance measurements and concepts from Ohm's Law.

Example Problem:

A resistor of resistance 1.5Ω is tested using a continuity function. If the multimeter shows a value of 1.5, does this indicate continuity?

Answer: Yes. The resistance is low, so continuity exists



10. EVALUATION

Objective Questions (WAEC/NECO Style):

Continuity testing is used to:

- A. Measure power
- B. Detect open circuits ✓
- C. Measure voltage
- D. Generate electric current

A good conductor when tested for continuity shows:

- A. High resistance
- B. No reading
- C. A beep sound ✓
- D. Spark

Theory Questions:

Define continuity testing and explain how it works.

Describe how a test lamp can be used to detect a broken wire.

Differentiate between continuity test and resistance measurement.

List five components that can be tested for continuity and the interpretation of results.

A student tests a switch using a multimeter and hears no sound. What could this imply? Suggest a remedy.



11. ASSIGNMENT



- (a) Sketch a circuit diagram showing a continuity tester connected to a wire and a battery.
- (b) Explain the step-by-step procedure for testing continuity in a car's headlamp wiring system.
- (c) In what ways is continuity testing useful in electronics manufacturing industries?

